

# n8n Workflow: Automate Ad Scraping, AI Rewriting, and Google Sheets Integration

This n8n workflow automates the following tasks:

1. **Scrape Ads:** Extract advertisements from Facebook and Instagram.
  2. **Extract Content:** Retrieve video and script content from the ads.
  3. **Convert Video to Audio:** Use CloudConvert to convert MP4 videos to MP3 format.
  4. **Transcribe Audio:** Convert audio content into text.
  5. **AI Rewriting:** Utilize AI to rewrite the ad scripts for enhanced engagement.
  6. **Store Results:** Upload the rewritten scripts and associated data to Google Sheets for tracking and analysis.
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## Prerequisites

### 1. CloudConvert Account & API Key

To convert MP4 videos to MP3:

1. Create a free account at [CloudConvert](#).
2. Navigate to the [API Key page](#).
3. Generate and copy your API key.

### 2. Apify Account & API Token

For advanced web scraping tasks:

1. Sign up for a free account at [Apify](#).
2. Go to the [Integrations page](#).
3. Generate and copy your API token.

### 3. Google Account with Google Sheets Access

Ensure you have a Google account with access to Google Sheets where the final data will be stored.

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## Setup Instructions

### 1. Configure Credentials in n8n

- **CloudConvert:**
  - In n8n, navigate to **Credentials**.
  - Create new credentials for **CloudConvert**:
    - **Name:** `CloudConvert API`
    - **API Key:** Paste your CloudConvert API key.
- **Apify:**
  - In n8n, navigate to **Credentials**.
  - Create new credentials for **Apify**:
    - **Name:** `Apify API`
    - **API Token:** Paste your Apify API token.
- **Google Sheets:**
  - In n8n, navigate to **Credentials**.

- Create new credentials for **Google Sheets**:
  - Follow the OAuth2 authentication process to grant n8n access to your Google Sheets.

## 2. Import the Workflow

- In n8n, click on **Import**.
  - Paste the workflow JSON or upload the workflow file.
  - Replace placeholder values with your actual data (e.g., API keys, sheet IDs).
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## Workflow Details

### Step 1: Scrape Ads from Facebook and Instagram

- **Method**: Utilize Apify Actors or Facebook Graph API to scrape ads.
- **Output**: List of ads with associated video URLs and scripts.

### Step 2: Download Video Content

- **Node**: HTTP Request
- **Function**: Download MP4 video files from the scraped ad URLs.

### Step 3: Convert MP4 to MP3

- **Node**: CloudConvert
- **Function**: Convert downloaded MP4 videos to MP3 format for transcription.

### Step 4: Transcribe Audio to Text

- **Node**: AI Service (e.g., OpenAI Whisper)

- **Function:** Transcribe MP3 audio files into text scripts.

## Step 5: Rewrite Scripts Using AI

- **Node:** AI Service (e.g., OpenAI GPT-4)
- **Function:** Rewrite transcribed scripts to enhance engagement and clarity.

## Step 6: Upload to Google Sheets

- **Node:** Google Sheets
  - **Function:** Append the rewritten scripts along with metadata (e.g., original ad URL, date) to a specified Google Sheet.
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## References

- **n8n Documentation:** <https://docs.n8n.io>
  - **CloudConvert API Documentation:** <https://cloudconvert.com/api/v2>
  - **Apify API Documentation:** <https://docs.apify.com/api/v2>
  - **Google Sheets API Documentation:** <https://developers.google.com/sheets/api>
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## Troubleshooting

- **Authentication Errors:**
  - Ensure all API keys and tokens are correctly entered and active.
  - Verify that the credentials have the necessary permissions.
- **Data Extraction Issues:**

- Check if the ad URLs are accessible and valid.
- Ensure that the scraping methods comply with platform policies.
- **File Conversion Failures:**
  - Confirm that the video files are in a supported format.
  - Check CloudConvert's service status for any outages.
- **AI Processing Errors:**
  - Review the input data for completeness and correctness.
  - Monitor API usage limits and quotas for the AI services used.